

EXPERIMENT 14

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Aim

To prepare an algorithm and test-case table, then write, compile, execute, and test a menu-driven C program using **switch-case** to check digit uniqueness and find the frequency of digits 0 to 9 in a positive integer.

Problem Statement

Input a positive integer and perform one operation based on the user choice: **(1)** check whether all digits are unique, or **(2)** find the frequency of each digit from 0 to 9.

Algorithm

1. Start and input a positive integer n.
2. If $n \leq 0$, display an error and stop.
3. Display the menu and input choice.
4. Create an integer array freq[10] and initialize every element to 0.
5. Use switch(choice).

Case 1: Copy n to temp. Repeatedly extract digit = temp % 10. If freq[digit] is already greater than 0, a repeated digit exists and the number is Not Unique. Otherwise increment freq[digit]. Continue until temp becomes 0. If no repetition occurs, display Unique.

Case 2: Copy n to temp. Extract every digit and increment freq[digit]. After processing all digits, display freq_0 through freq_9.

Default: Display Invalid choice.

6. Stop.

Test Case Table

Test	Number	Choice	Expected Result
1	12345	1	Unique
2	12234	1	Not Unique
3	98760	1	Unique
4	112333	2	0:0, 1:2, 2:1, 3:3, others:0
5	10021	2	0:2, 1:2, 2:1, others:0
6	5555	2	5:4, all other digits:0
7	123	3	Invalid choice

C Program

```
#include <stdio.h>

int main()
{
    int n, temp, choice, digit, i;
    int freq[10] = {0};
    int unique = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Check whether all digits are unique\n");
    printf("2. Find frequency of each digit from 0 to 9\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            temp = n;
            while (temp > 0) {
                digit = temp % 10;
                if (freq[digit] > 0) {
                    unique = 0;
                    break;
                }
                freq[digit]++;
                temp /= 10;
            }
            if (unique)
                printf("Unique\n");
            else
                printf("Not Unique\n");
            break;

        case 2:
            temp = n;
            while (temp > 0) {
                digit = temp % 10;
                freq[digit]++;
                temp /= 10;
            }
            for (i = 0; i <= 9; i++)
                printf("freq_%d = %d\n", i, freq[i]);
            break;

        default:
            printf("Invalid choice.\n");
    }
    return 0;
}
```

Compilation and Execution (Linux/GCC)

Save as: experiment14.c

Compile: gcc experiment14.c -o experiment14

Execute: ./experiment14

Sample Executions and Testing

Test 1: n = 12345, choice = 1 -> **Unique**. Result: **PASS**

Test 2: n = 12234, choice = 1 -> **Not Unique**. Result: **PASS**

Test 3: n = 112333, choice = 2 -> freq_0=0, freq_1=2, freq_2=1, freq_3=3, freq_4=0, freq_5=0, freq_6=0,

freq_7=0, freq_8=0, freq_9=0. Result: **PASS**

Result

The menu-driven C program was successfully prepared using switch-case, compiled, executed, and tested. It correctly checks whether all digits are unique and calculates the frequency of every digit from 0 to 9.